

SQL Assignments - Day 2 Set 2

Common Instructions

- For character data type either char or varchar should be taken as per the case and appropriate size should be given.
- For numeric data type select appropriate sub data type within numeric family.
- Enter at least 5 records and ensure the validity of data types, size and constraints.
- If the table with the same name already exists in your schema, then do not drop it. Provide some different name for your table.

- Create the table student_master with the following specifications

Column Name	Data type	Constraints and other details
Rollno	Numeric	Primary Key
Name	Character	Cannot be empty
City	Character	By default, it should be Mumbai.
Marks	Numeric	Cannot be less than 0 and cannot exceed 100
Invoice_No	Character	Cannot repeat, but can be blank

Query: create table student_master(Rollno INT PRIMARY KEY, Name char(20) NOT NULL, City char(15) DEFAULT 'Mumbai', Marks INT check(Marks BETWEEN 0 and 100), Invoice_No char(20) UNIQUE NOT NULL);

Insert into student_master values(10,'Arpit','Buldhana',99,101),

-> (11,'Warade','Buldhana',99,102),

-> (12,'Patil','Buldhana',94,103),

-> (13,'Patel','Buldhana',80,104),

-> (14,'IevaPatel','Buldhana',95,105);

Query OK, 5 rows affected (0.09 sec)

Records: 5 Duplicates: 0 Warnings: 0

- Create the table certificate_details with the following specifications

Column Name	Data type	Constraints and other details
Rollno	Numeric	Taken from student_details table's rollno.
Date Of Certificate	Date	Cannot be empty
Course	Character	Can be Oracle or Power BI or Azure Data Factory

Query: create table certificate_details(Rollno INT REFERENCES STUDENT_MASTER(RollNo), Date_of_Certificates Date NOT NULL, course char(15) check(course IN('Oracle', 'Power BI', 'Azure Data factory')));

Insert into certificate_details values(10,'2025-08-03','Oracle'),

-> (11,'2024-08-03','Power BI'),

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-> (12,'2024-03-21','Oracle'),
-> (13,'2023-12-02','Oracle'),
-> (14,'2024-11-07','Power BI');

3. Create the table First_Attempt_Marks with the following specifications

Column Name	Data type	Constraints and other details
Rollno	Numeric	Rollno and Subject should make a composite primary key
Subject	Character	
Marks	Numeric	Cannot be empty

Query: create table First_Attempt_Marks(RollNo INT, Subject char(10),Marks INT NOT NULL, primary key(RollNo,Subject));

Insert into first_attempt_marks values(10,'dbms',78),

-> (11,'os',87),
-> (12,'cpp',99),
-> (13,'Java',85),
-> (14,'Opps',98);

4. Create the table First_Attempt_Grades with the following specifications

Column Name	Data type	Constraints and other details
Rollno	Numeric	
Subject	Character	
Grade	Character	Should be either A or B
		Rollno and Subjects should be taken from First_Attempt_Marks

Query: create table First_Attempt_Grades(RollNo INT, Subject char(10),Foreign key(RollNo,Subject) REFERENCES First_Attempt_Marks(RollNo,Subject),Grade char(2));

Insert into first_attempt_grades values(10,'dbms','A'),

-> (11,'os','A+'),
-> (12,'cpp','O'),
-> (13,'Java','B+'),
-> (14,'Opps','B');

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5. Create the table cost_analysis with the following specifications

Column Name	Data type	Constraints and other details
Company ID	Numeric	Primary Key
Company Name	Character	Cannot be empty
Export Cost	Numeric	
Import Cost	Numeric	
<i>Import Cost should always be greater the or equal to 40% of the export cost</i>		

Query: create table cost_analysis(CompanyId INT PRIMARY KEY, companyName char(30) NOT NULL, Export_Cost INT, ImportCost INT CHECK(ImportCost >= 0.4 * Export_Cost));

insert into cost_analysis values(11,'TCS',75, 140),
-> (12,'zensar', 87, 160),
-> (13,'Google',75,140),
-> (14,'Amazon',67,130),
-> (15,'oracle',77,160);

6. Create the table Salesman_Data with the following specifications

Column Name	Data type	Constraints and other details
Salesman ID	Numeric	Primary Key
Salesman Name	Character	Cannot be blank
Area Code	Numeric	Cannot repeat
Sales Amt	Numeric	
Profit Amt	Numeric	
<i>Either Sales Amt should be above 5000 OR Profit Amt should be above 700.</i>		

Query: create table salesman_data(SalesmanID INT PRIMARY KEY, salesmanName char(20) NOT NULL, Area_Code INT unique, Sales_Amt INT, Profit_Amt INT, CHECK(Sales_Amt>5000 or Profit_Amt>700));

insert into salesman_data values(11,'Arpit',41104, 5500,200),
-> (12,'XYZ',42104, 600,800),
-> (13,'ABC',42004, 5000,1000),
-> (14,'PQR',42014, 6000,1000),
-> (15,'XQR',41014, 4000,9000);